

University of Mumbai

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विद्याविषयक प्राधिकरणे
सभा आणि सेवा विभाग(ए.ए.एम.एस)
कम नं. १२८ एम.जी.रोड, फोर्ट,
मुंबई - ४०० ०३२
टेलिफोन नं - ०२२ - ६८३२००३३

(नॅक पुनर्मूल्यांकनाद्वारे ३.६५ (सी.जी.पी.ए.) सह अ++ श्रेणी
विद्यापीठ अनुदान आयोगाद्वारे श्रेणी १ विद्यापीठ दर्जा)

क.वि.प्रा.स.से./आयसीडी/२०२५-२६/३७

दिनांक : २७ मे, २०२५

परिपत्रक:-

सर्व प्राचार्य/संचालक, संलग्नित महाविद्यालये/संस्था, विद्यापीठ शैक्षणिक विभागांचे संचालक/ विभाग प्रमुख यांना कळविण्यात येते की, राष्ट्रीय शैक्षणिक धोरण २०२० च्या अमलबजावणीच्या अनुषंगाने शैक्षणिक वर्ष २०२५-२६ पासून पदवी व पदव्युत्तर अभ्यासक्रम विद्यापरिषदेच्या दिनांक २८ मार्च २०२५ व २० मे, २०२५ च्या बैठकीमध्ये मंजूर झालेले सर्व अभ्यासक्रम मुंबई विद्यापीठाच्या www.mu.ac.in या संकेत स्थळावर NEP २०२० या टॅब वर उपलब्ध करण्यात आलेले आहेत.

मुंबई - ४०० ०३२
२७ मे, २०२५


(डॉ. प्रसाद कारडे)
कुलसचिव

क.वि.प्रा.स.से.वि/आयसीडी/२०२५-२६/३७ दिनांक : २७ मे, २०२५
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As Per NEP 2020

University of Mumbai



Revised Syllabus for

Title of the U.G. program

B. Sc. (Physics)

Semester – Sem I & II

(Scheme – III)

Ref: GR dated 20th April, 2023 for Credit Structure of UG

(With effect from the academic year 2025-26)

University of Mumbai



(As per NEP 2020)

Sr. No.	Heading	Particulars
1	Title of program O: _____	B.Sc. (Physics)
2	Exit Degree	U.G. Certificate in Physics
3	Scheme of Examination R: _____	NEP 40% Internal 60% External, Semester End Examination Individual Passing in Internal and External Examination
4	Standards of Passing R: _____	40%
5	Credit Structure Sem I & II R: SU – 520 A (III) R: SU – 520 B (III)	Attached herewith
6	Semesters	Sem. I & II
7	Program Academic Level	5.00
8	Pattern	Semester
9	Status	Revised
10	To be implemented from Academic Year	2025-26

Sd/-

Sign of the BOS Chairman
Dr. T.N. Ghorude
Board of Studies in
Physics

Sd/-

Sign of the
Offg. Associate Dean
Dr. Madhav R. Rajwade
Faculty of Science &
Technology

Sd/-

Sign of the
Offg. Dean
Prof. Shivram S. Garje
Faculty of Science &
Technology

Preamble

1) Introduction:

The revised syllabus in Physics, as per credit based system for the First Year B. Sc. course will be implemented from the academic year 2024 – 2025.

2) Aims and Objectives

The systematic and planned curricula from these courses shall motivate and encourage learners to understand basic concepts of Physics.

Objectives:

- To develop analytical abilities towards real world problems
- To familiarize with current and recent scientific and technological developments
- To enrich knowledge through problem solving, hands on activities, study visits, projects etc.
- To acquire knowledge of fundamental optics.

3) Learning Outcomes

On successful completion of this course students will be able to:

Understand the Newton's laws of motion, friction, work, energy and able to solve problems using them. Understand the mechanics of multi-particle system using concepts of center of mass and Conservation laws.

Study the mechanics of undamped (simple harmonic motion, uniform circular motion) and damped oscillations (Forced oscillations, two body oscillation) and their implementation in physical applications such as torsional, compound, and simple pendulums.

Understand AC circuit theory in case of pure resistance, inductance, capacitance and series combinations of LR, CR and LCR circuits, the working of AC bridges such as Maxwell's Inductance bridge, De Sauty's bridge, Wien bridge.

Comprehend circuit theorems (Ohm's law, Kirchoff's laws, Thevenin's, Norton's, and Maximum Power Transfer theorems). Also, understand magnetic properties of matter, concepts of magnetic permeability, magnetic forces, magnetic field, magnetization, Biot-Savart's law.

Study the Lens maker's equation, Newton's lens equation and principal foci positions.

Also, understand Lateral, Longitudinal and Angular magnification, Equivalent focal length and power of two thin lenses, Concept of cardinal points and their significance.

Comprehend Spherical aberration & reduction, chromatic aberration & reduction. Also, understand Fresnel and Fraunhofer type of diffraction and Fraunhofer diffraction pattern due to a single slit and double slit, Michelson's Interferometer and its Application, Polarization and types of Polarization.

Comprehend the concepts of DC power supply and familiarize with diode and Zener diode circuits and its applications, Methods of Transistor Biasing, Base Resistor or fixed bias, Emitter Bias and Voltage Divider Bias Methods.

Understand DC transistors Biasing and α , β (dc and ac) gain, Inherent Variations of transistor Parameters and Stabilization, Derived Gates NAND,NOR and Ex-OR gate, Including their symbols and truth table.

Apply the knowledge to design logical circuit using basic gates and its applications.

Students must be able to work through problems pertaining to the topics covered in the syllabus.

5) Credit Structure of the Program (Sem I, II, III, IV, V & VI)

Under Graduate Certificate in Physics

Credit Structure (Sem. I & II)

	R: _____ A									
Level	Semester	major		Minor	O E	VSC , (VSEC)	AEC, VEC, IKS, SEC	OJT, FP, CEP, CC, RP	Cum. Cr./Sem.	Degree/ Cum. Cr.
		Mandatory	Electives							
4.5	I	MJ – 1:Introduction to Mechanics (2) MJ – 2: Basic Electricity & Magnetism (2) MJP – 1: Physics practical – I (2)	---	-	2	VSC1: Laboratory Equipment Maintenance (2)	SEC1: Basic Instrumentation skill and Basic mathematical skills for Physics (2)	2	22	UG Certificate 44
	R: _____ B									
	II	MJ – 3: Optics(2) MJ – 4:Fundamentals of Electronics (2) MJP – 2 : Physics Practical – II (2)	---	2	2	VSC2: PCB Making (2)	SEC2: ICT tools for Physics (2)	2	22	
	Cum Cr.	12	---	2	8	4+4	4+4+2	4	44	
Exit option: Award of UG Certificate in Major with 40-44 credits and an additional 4 credits core NSQF course/ Internship OR Continue with Major and Minor										

Under Graduate Diploma in Physics

Credit Structure (Sem. III & IV)

	R:_____C									
5.0	III	MJ – 5:Mathematical Methods for Physics (2) MJ – 6: Electrodynamics –I (2) MJP-3: Physics Major Practical – III (2) MJP-4 : Physics Major Practical – IV (2)	- - -		4	VSC3: Computer Maintenance (2)	SEC3:		22	UG Diploma 88
	R:_____D									
	IV	MJ – 7: Introduction to Quantum Mechanics and Modern Physics (2) MJ – 8: Advanced Electronics –I (2) MJP- 5: Physics Major Practical – V (2) MJP-6:Physics Major Practical – VI (2)	- - -		4	VSC4: Battery and Inverter Testing & Maintenance (2)	SEC4:		22	
	Cum Cr.	28		10	12	6+6	8+4+2	8+4	88	
Exit option; Award of UG Diploma in Major and Minor with 80-88 credits and an additional 4 credits core NSQF course/ Internship OR Continue with Major and Minor										

B.Sc. (Physics)

Credit Structure (Sem. V & VI)

	R:_____E									
5.5	V	MJ – 9: Thermal & Statistical Physics (2) MJ – 10: Solid State Physics (2) MJ – 11: Atomic & Molecular Physics (2) MJP- 7: Physics Major Practical – VII (2) MJP- 8: Physics Major Practical – VIII (2)	EL – I: Electronic Instrumentation –I (2) or Microcontrollers (2) or Numerical Techniques - I (2) ELP – I: Practical of EL I (2)		4	VSC5: Study of Solar Panel and its Installation (2)	SEC5:		22	UG Degree 132
	R:_____F									
	VI	MJ – 12: Classical Physics (2) MJ – 13: Nuclear Physics (2) MJ – 14: Special Theory of Relativity (2) MJP – 9: Physics Major Practical – IX (2) MJP – 10: Physics Major Practical – X (2)	EL –II: Electronic Instrumentation – II (2) or Numerical Techniques – II (2) or Python Programming – I (2) ELP – II: Practical of EL II (2)		4	VSC6: Electrical Energy Auditing (2)	SEC6:		22	
	Cum Cr.	48	8	18	12	8+6	8+4+2	8+6+4	132	
Exit option: Award of UG Degree in Major with 132 credits OR Continue with Major and Minor										

[Abbreviation - OE – Open Electives, VSC – Vocation Skill Course, SEC –Skill Enhancement Course, (VSEC), AEC – Ability Enhancement Course, VEC – Value Education Course, IKS – Indian Knowledge System, OJT – on Job Training, FP – Field Project, CEP – Community Engagement Project, CC – Co-Curricular, RP – Research Project]

Sem. - I

Syllabus
B.Sc. (Physics)
(Sem.- I)

Name of the Course: F.Y.B.Sc. SEM-I - Paper I – Introduction to Mechanics

Sr.No.	Heading	Particulars
1	Description the course : Including but Not limited to:	Introduction, relevance, Usefulness, Application, interest, connection with other courses, demand in the industry, job prospects etc.
2	Vertical :	Major/Minor/Open Elective /Skill Enhancement / Ability Enhancement/Indian Knowledge System
3	Type :	Theory / Practical
4	Credits :	2 credits (1 credit = 15 Hours for Theory or 30 Hours of Practical work in a semester)
5	Hours Allotted :	30 Hours
6	Marks Allotted:	50 Marks
7	Course Objectives (CO): After successful completion of this course students will be able to: CO 1. Explain Newton's laws of motion, friction, work, energy and able to solve problems using them. CO 2. Learn the mechanics of multi-particle system using concepts of center of mass and conservation laws. CO 3. Study the mechanics of undamped (simple harmonic motion, uniform circular motion) and damped oscillations (Forced oscillations, two body oscillation) CO 4. Describe qualitatively how undamped and damped oscillations are implemented in physical problems such as torsional, compound, and simple pendulums. CO 5. Demonstrate quantitative problem solving skills in all the topics covered in the syllabus.	
8	Course Outcomes (OC): After successful completion of this course the learner will be able to: OC 1. Understand Newton's laws of motion, friction, work, energy and able to solve problems using them.	

	OC 2. Comprehend Work and Energy equivalence and its applications through suitable numerical. OC 3. Understand mechanics of multi-particle system using concepts of center of mass and conservation laws. OC 4. Understand mechanics of undamped/ (simple harmonic motion, uniform circular motion) and damped oscillations OC 5. Understand how undamped and damped oscillations are implemented in physical problems OC 6. Demonstrate quantitative problem solving skills in all the topics covered
9	Modules:-Paper 1 – Introduction to Mechanics (30 Hrs) UNIT- I (15 Hrs) 1. Newton’s Laws of Motion: Newton’s first, second and third laws of motion, interpretation and applications, pseudo forces, inertial and non-inertial frames of reference Worked out examples (with friction present). (HCV: 5.1 to 5.5) 2. Friction: Advantages & disadvantages of friction in daily life, Friction as the component of Contact force, Kinetic Friction, Static friction, laws of friction, Understanding friction at atomic level. (HCV: 6.1 to 6.5) 3. Work and Energy: Kinetic Energy, Work and Work-energy theorem, Potential Energy, Conservative and Non-Conservative Forces, Different forms of Energy: Mass Energy Equivalence Worked out Examples. (HCV: 8.1, 8.2, 8.5, 8.6, 8.11) UNIT- II (15 Hrs) 1.Many Particles System, Centre of Mass of solid objects, Conservation of momentum in a system of particle, Angular momentum of a particle and system of particle, conservation of angular momentum. (RH: 7.3, 7.4, 7.5, 10.1, 10.2, 10.4) Oscillations: The Simple Harmonic Oscillator, Relation between Simple Harmonic Motion and Uniform Circular Motion, Damped Harmonic Motion, Forced Oscillations and Resonance, Two Body Oscillations. RH:17.2, 17.6, 17.7, 17.8, 17.9 Examples of Simple Harmonic oscillations: Simple Pendulum, Simple Pendulum, Torsional Pendulum and Compound pendulum (Qualitative study) HP: 9.1.1(1,3,4)

10	Text Books		
11	Reference Books 1. HCV: H.C. Verma, Concepts of Physics-Part I (Second Reprint of 2020) BharatiBhavan Publishers and Distributers 2. RH: Resnick and Halliday: Physics – I , 5 th Edition. 3. Mechanics – H. S. Hans and S. P. Puri, Tata McGraw Hill (2nd ED.).		
12	Internal Continuous Assessment: 40% (20Marks)	Semester End Examination: 60% (30 Marks)	
13	Continuous Evaluation through: Quizzes, Class Tests, presentation, project, role play, creative writing, assignment etc.(at least 3)		
14	Format of Question Paper: 30 Marks Duration: One Hour		
	Unit -I (15Marks)	Q:1 A) Attempt any Two 10Marks i) Theory ii) Theory iii) Theory iv) Theory B) Attempt any One 05 Marks i) Problem ii) Problrm	
	Unit -II (15Marks)	Q:2 A) Attempt any Two 10Marks i) Theory ii) Theory iii) Theory iv) Theory B) Attempt any One 05 Marks i) Problem ii) Problrm	

Name of the Course: F.Y.B.Sc. SEM-I Paper – II: Basic Electricity and Magnetism

Sr. No.	Heading	Particulars
1	Description the course : Including but Not limited to:	Introduction, relevance, Usefulness, Application, interest, connection with other courses, demand in the industry, job prospects etc.
2	Vertical :	Major/Minor/Open Elective /Skill Enhancement / Ability Enhancement/Indian Knowledge System
3	Type :	Theory / Practical
4	Credits :	2 credits (1 credit = 15 Hours for Theory or 30 Hours of Practical work in a semester)
5	Hours Allotted :	30 Hours
6	Marks Allotted:	50 Marks
7	Course Objectives (CO): After successful completion of this course students will be able to: CO 1. Explain the AC circuit theory in case of pure resistance, inductance, capacitance and series combinations of LR, CR and LCR circuits. CO 2. Apply the knowledge of AC circuit theory to understand the working of AC bridges such as Maxwell's inductance bridge, De Sauty's bridge, Wien bridge CO 3. Explain basic circuit theorems (Ohm's law, Kirchoff's laws, Thevenin's, Norton's, and Maximum Power Transfer theorems). CO 4. Describe magnetic properties of matter, concepts of magnetic permeability, magnetic forces, magnetic field, magnetization, Biot Savarts law. CO 5. Solve numerical based on the topics that are covered in the syllabus.	
8	Course Outcomes (OC): After successful completion of this course students will be able to: OC 1. Understand AC circuit theory in case of pure resistance, inductance, capacitance and series combinations of LR, CR and LCR circuits.	

	OC 2. Understand the working of AC bridges such as Maxwell's inductance bridge, De Sauty's bridge, Wien bridge OC 3. Comprehend circuit theorems (Ohm's law, Kirchoff's laws, Thevenin's, Norton's, and Maximum Power Transfer theorems). OC 4. Understand magnetic properties of matter, concepts of magnetic permeability, magnetic forces, magnetic field, magnetization, Biot-Savarts law. OC 5. Students learn to apply their knowledge to solve problems related to the topics that are covered in the syllabus.
9	Paper 2 – Electricity and Magnetism (30 Hrs) UNIT-I (15 Hrs) 1. Alternating current theory: (Review: Concept of L, R, and C) AC circuit containing pure R, pure L and pure C, representation of sinusoids by complex numbers, Series L-R, C-R and LCR circuits, Resonance in LCR circuit (series), Q- Factor. (TT: 11.29, 11.30, 11.32, 12.5, 12.6, 13.1, 13.7, 13.9, 13.10, 13.11, 13.12, 13.13, 13.14, 13.17). 2. AC bridges: General AC Bridge, Maxwell's Inductance Bridge, Maxwell's L/C Bridge, De Sauty Bridge, Wien Bridge. (Bridge diagram, balancing condition derivation, applications). (TT: 16.1, 16.2, 16.3, 16.9, 16.11, 16.12). UNIT-II (15 Hrs) 1.Circuit Theorems: (Review: Ohm's law, Kirchhoff's laws) Ideal Current and Voltage Sources, Thevenin's Theorem, Norton's Theorem, Maximum Power Transfer Theorem. Problems related to circuit analysis using the above theorems. (TT: 2.15, 2.16, 2.18, 2.25, 2.30) (Coulomb's Law, The Electric Field for Review) Magnetic properties of matter: Introduction, Magnetic Permeability, Magnetization (Chapter 9: 1, 2, 3) SOP: Solid State Physics, S.O Pillai (5th Edition), New Age International Limited. Magnetostatics: Magnetic Fields, Magnetic forces, Currents The Biot-Savart Law: Steady Currents, The Magnetic Field of a Steady Current (DJG: 5.1.1, 5.1.2, 5.1.3, 5.2, 5.2.1, 5.2.2)
10	Text Books
11	References:

	1. TT: B.L, Theraja and A.K. Theraja, A Textbook of Electrical Technology Vol. I, S. Chand Publication 2. DJG: Introduction to Electrodynamics 3rd Edn by D. Griffith 3. BS: Mechanics and Electrodynamics Rev Edn. 2005, by Brijlal and Subramanayan and Jeevan Seshan	
12	Internal Continuous Assessment: 40% (20 Marks)	Semester End Examination: 60% (30 Marks)
13	Continuous Evaluation through: Quizzes, Class Tests, presentation, project, role play, creative writing, assignment etc.(at least 3)	
14	Format of Question Paper: 30 Marks Duration One Hour	
	Unit -I (15Marks)	Q:1 A) Attempt any Two 10Marks i) Theory ii) Theory iii) Theory iv) Theory B) Attempt any One 05 Marks i) Problem ii) Problem
	Unit -II (15Marks)	Q:2 A) Attempt any Two 10Marks i) Theory ii) Theory iii) Theory iv) Theory B) Attempt any One 05 Marks i) Problem ii) Problem

SEMESTER I

PHYSICS PRACTICAL COURSE –USPHP1

INSTRUCTIONS:

- 1) All the measurements and readings should be written with proper units in SI system only.
- 2) After completing all the required number of experiments in the semester and recording them in journal, student will have to get their journal certified and produce the certified journal at the time of practical examination.
- 3) While evaluating practical, weightage should be given to circuit/ray diagram, observations, tabular representation, experimental skills and procedure, graph, calculation and result.
- 4) Skill of doing the experiment and understanding physics concepts should be more important than the accuracy of final result.

Note: Exemption of two experiments from section A and / or B and / or C may be given if student carries out any one of the following activities.

- Collect the information of at least five Physicists with their work or any three events on physics, report that in journal.
- Execute a mini project to the satisfaction of teacher in-charge of practical.
- Participate in a study tour or visit & submit a study tour report.
- For practical examinations, the learner will be examined in ONE experiments (one from any group).
- One Experiment will be of **TWO** lecture hours' duration.
- A Minimum 4 from each group and in all minimum 8 experiments must be reported in journal.
- All the skill experiments are required to be completed compulsorily. Students are required to report all these experiments in the journal. Evaluation in viva voce will be based on regular experiments and skill experiments.

A learner will be allowed to appear for the semester and practical examination only if he submits a certified journal of Physics or a certificate that the learner has completed the practical course of Physics Semester I as per the minimum requirements.

A. Regular Experiment:

Sr No	Name of the Experiment
GROUP A	
1	Torsional Oscillation: To determine modulus of rigidity η of a material of wire by Torsional oscillations
2	Bifilar Pendulum: Determination of moment of inertia of rectangular and cylindrical bar about an axis passing through its centre of gravity
3	Moment of inertial of Flywheel
4	Young's Modulus of a wire material by method of vibrations
5	Bar Pendulum- determination of g
6	LDR Characteristics: To study the dependence of LDR resistance on intensity of light
GROUP B	
7	Frequency of AC Mains: To determine frequency of AC mains (Sonometer wire)
8	To study Thermistor characteristics: Resistance Vs Temperature
9	To determine capacitance in AC circuits using R and C
10	To determine Inductance in AC circuits using L and C
11	To determine the horizontal component of Earth's magnetic field(H) in the laboratory using deflection and vibration magnetometer
12	To determine the self-inductance of a coil with Anderson's Bridge
GROUP C: Skill Experiment	
1	Use of Vernier Callipers, Micrometer Screw Gauge and Travelling Microscope
2	Graph plotting (Plot BE/A verses A graph for 30 atoms, Plot Packing Fraction graph for 30 atoms)
3	Spectrometer: Schuster's Method
4	To determine the Resistance & Capacitance using Color code/Number & verify using Multimeter (Analog/Digital).
5	Use of digital multimeter
6	Absolute and relative error calculation

Note: Minimum **8** experiments (Four From each group) and **4** Skill experiments should be completed and reported in the journal, in the first semester. **Certified Journal is a must**, to be eligible to appear

for the semester end practical examination.

Semester End Practical Examination: 30 Marks External & 20 Marks Internal

Scheme of Examination: 30 Marks Duration: TWO Hours

A candidate will be allowed to appear for the semester end practical examination only if the candidate submits a certified journal at the time of practical examination of the semester or a certificate from the Head of the Department /Institute to the effect that the candidate has completed the practical course of that semester of F.Y.B.Sc. Physics as per the minimum requirement. The duration of the practical examination will be two hours. There will be one experiment (from any group) through which the candidate will be examined in practical. The questions on slips for the same should be framed in such a way that candidate will be able to complete the task and should be evaluated for its skill and understanding of physics.

F.Y.B. Sc. Physics Evaluation pattern and Question paper pattern for Semester End Practical Examination of Major Course

Internal Continuous Assessment: 40% (20 Marks)	Duration of semester End examination (Internal)
Viva/ assignment/ objective question test (15 Marks), Overall performance (5 Marks) = 20 Marks	1 Hr

External Semester End Examination: 60% (30 Marks)	Duration of semester End examination (External)
As per paper pattern	2 Hr

Continuous Evaluation through: Practical

Format of Question Paper:**Major Practical: 2 credits****Practical of 2 credits, Duration: 60 Hrs , Total marks: 50****(30 Marks for External Practical Exam + 20 Marks for Internal Practical Exam.)****Paper Pattern for 30 marks External Semester End Practical Examination:****Duration** - This examination shall be of **Two hours** duration in laboratory

Sr. No.		Total 30 Marks
1	Experimental performance	20 Marks
2	Journal	05 Marks
3	VIVA	05 Marks

Paper Pattern for 20 marks Internal Semester End Practical Examination:Duration - This examination shall be of **One hour** duration in laboratory

Sr. No.		Total 20 Marks
1	Practical/Theoretical/ Combined Performance	15 Marks
2	Overall Performance	05 Marks

Title: - Laboratory Equipment Maintenance

Course Objectives

After successful completion of this course students will be able to:

1. learn to identify and testing of electronic components.
2. Detecting faults in circuits and troubleshooting them.
3. Repair of different types of laboratory power supplies.
4. Learn Maintenance of basic laboratory equipment's.

Course Objectives practical's:

After successful completion of this course students will be able to:

1. Identify different types of wires, power cables, probes, fuses etc.
2. Identify, measure and testing of various types passive components and semiconductor devices.
3. Understand Soldering and desoldering given circuit on PCB and testing
4. Learn about of A.C & D.C. power supply & troubleshooting

Course Outcomes

After successful completion of this course students will be able to:

1. Gain knowledge of the different electronic components
2. Establish a basis for honing their practical electronics skills
3. Develop a lucrative and advanced profession in electronics.
4. Understand maintenance of basic laboratory equipment's

Course Outcomes Practicals:

After successful completion of this course students will be able to:

1. Comprehend to identify and testing of electronic components.
2. Understand to identify faulty components and troubleshoot circuits
3. Acquire skills of circuit Soldering and desoldering of electronic components in a given circuit
4. Comprehend hands on experience in handling and maintaining laboratory/electronic equipment.

Unit-I (15 Periods)

Testing of Passive Electronic Components using Digital Multimeter (DMM)

Passive components - resistors, capacitors, inductors, failures in fixed resistors, testing of resistors, variable resistors, variable resistor as potentiometer, measuring resistors using color codes.

Testing of various types of capacitors & inductors.

Unit-II (15 Periods)

Testing of semiconductor devices, soldering and troubleshooting.

Semiconductor devices: diode, bipolar junction transistors etc, causes & types of failures in semiconductor devices.

Basics of soldering: soldering alloy, soldering iron, soldering & desoldering, dry soldering and good contact.

Laboratory Power supplies – basic testing and troubleshooting.

List of Experiment

- 1.To identify different types of Wires, power cables, probes, fuses used in the laboratory and check their continuity using DMM
- 2.Identification of various types passive components - resistors, capacitors and inductors used in laboratory.
- 3.Measurement of resistors using colour codes & DMM and testing of capacitors and inductors.
- 4.Identification of various types of semiconductor devices: diode, bipolar junction transistors, Field effect transistors etc.
- 5.Testing of semiconductor devices: diode, bipolar junction transistors, Field effect transistors etc.
- 6.Soldering and desoldering of electronic components in a given circuit
- 7.Mounting simple circuit on PCB and testing.
- 8.Troubleshooting a given circuit.
- 9.Study of a.c power supply & troubleshooting
- 10.Study of d.c power supply and troubleshooting

Reference Books

- 1.Text book of Electrical Technology, by B.L. Theraja and A.K. Theraja
2. Modern Electronic Equipment: Troubleshooting, Repair and Maintenance by Khandpur, TMH 2006
3. Electronic Instruments and Systems: Principles, Maintenance and Troubleshooting by R. G. Gupta
Tata McGraw Hill Edition 2001
4. Student Reference Manual for Electronic Instrumentation Laboratories by Stanley Wolf, and
Richard F. M. Smith, Prentice Hall of India Pvt. Ltd. New Delhi
5. Consumer Electronics by S. P. Bali, Pearson
6. Troubleshooting and Maintenance of Electronic Equipment by K. Sudeep Singh

Reference Books Practical's:

1. Modern Electronic Equipment: Troubleshooting, Repair and Maintenance by Khandpur, TMH 2006

2. Student Reference Manual for Electronic Instrumentation Laboratories by Stanley Wolf, and Richard F. M. Smith, Prentice Hall of India Pvt. Ltd. New Delhi
3. Consumer Electronics by S. P. Bali, Pearson
4. Troubleshooting and Maintenance of Electronic Equipment by K. Sudeep Singh

Internal Continuous Assessment: 40% (20 Marks)	Semester End Examination: 60% (30 Marks)
Continuous Evaluation through: Practical's	

Format of Question Paper: 30 Marks Duration: One Hour

VSC: 2 credits

VSC of 2 credits, Duration: 60 Hrs , Total marks: 50

(30 Marks for Theory paper + 20 Marks for Practical Exam.)

Semester End Theory Examination: 60% (30 Marks)	Internal Semester End Practical Examination: 40 % (20 Marks)
As per paper pattern attached	As per practical exam. pattern attached

Theory Paper Pattern for 30 marks Semester End Theory Examination:

1. Duration - These examinations shall be of **one hours** duration

Duration: 1 Hrs

Total Marks -30

Unit -I (15Marks)	Q:1 A) Attempt any Two i) Theory ii) Theory iii) Theory iv) Theory B) Attempt any One i) Problem ii) Problem	10Marks 05 Marks
Unit -II (15Marks)	Q:2 A) Attempt any Two i) Theory ii) Theory	10Marks

	iii) Theory iv) Theory B) Attempt any One i) Problem ii) Problem	05 Marks
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Internal Practical Examination Pattern for 20 marks Semester End Examination:

1. Duration - These examinations shall be of **Two hours** duration in laboratory

Sr. No.		Total 20 Marks
1	One Experiment	15 Marks
2	Certified Journal	3 Marks
3	Vi-va	2 Marks

Physics (Skill Enhancement Course) - 2 credits

Course code: USPHSEC1

Title: - Basic Instrumentation skills and Basic Mathematical Skills for Physics

Sr. No.	Course Objective	Course Outcome
1.	Generate awareness among students about handling different laboratory instruments scientifically.	Accomplish desired skills to handle different laboratory instruments scientifically.
2.	Develop concepts of accuracy precision, resolution, range and errors/uncertainty in measurement.	Acquire knowledge about precision and accuracy in measurements.
3.	Understand various types of electronic components and devices so as to construct simple circuits	Develop basic electronic circuit using different techniques.
4.	Expose students to systematic of scientific calculator.	Develop confidence to use scientific calculator systematically.
5.	Illustrate necessary mathematical concepts to develop corresponding skills	Apply mathematical tools to understand theoretical concepts of physics.
6.	Develop the problem solving among learners	Demonstrate problem-solving skills for all the topics covered.

This course is designed for learners to get exposure with various aspects of instruments and their usage through hands-on mode.

Unit 1: Basic Instrumentation skills

(30 Lectures)

- 1.1. Study and use of Vernier Callipers & Micrometer Screw Gauge
- 1.2. Study and use of Travelling Microscope.
- 1.3. Study and use of Spectrometer.
- 1.4. Study and determination of Focal length of a convex lens.
- 1.5. Fundamentals of Electronic Component- Resistor, potentiometer, Capacitor, Inductor, Diode, Transistor, LED, Zener diode.
- 1.6. Use of Multimeter (analog, Digital), Voltmeter, Ammeter.
- 1.7. Use of CRO - voltage (AC, DC), Frequency and Phase measurement.
- 1.8. Different types of power supplies.
- 1.9. Logic gate IC Testing.
- 1.10. Soldering electronic circuits.
- 1.11. Study and use of Thevenin's theorem.
- 1.12. Building Electronic Circuits using Breadboard.
- 1.13. Graph plotting - plotting of graphs and finding slope and intercept.
- 1.14. Graph plotting - semi log graph paper.
- 1.15. Theory of errors and calculating-Error for given data

Unit 2: Basic Mathematical Skills

(30 Lectures)

- 2.1 Use of scientific Calculator
- 2.2 Basic trigonometry and its applications in physics
- 2.3 Concept of logarithm and its application to Physics
- 2.4 Concept of derivatives and its application to Physics

2.5 Concept of integration and its application to Physics

2.6 Concept of differential equations(1st order) and its application to Physics

F.Y.B. Sc. Physics Evaluation pattern and Question paper pattern for Semester End (SEC) Practical Examination

Internal Continuous Assessment: 40% (20 Marks)	Duration of semester End examination (Internal)
Viva/ assignment/ objective question test (15 Marks), Overall performance (5 Marks) = 20 Marks	1 Hr

External Semester End Examination: 60% (30 Marks)	Duration of semester End examination (External)
As per paper pattern	2 Hr

Continuous Evaluation through: Practical

Format of Question Paper:

Major Practical: 2 credits

Practical of 2 credits, Duration: 60 Hrs , Total marks: 50

(30 Marks for External Practical Exam + 20 Marks for Internal Practical Exam.)

Paper Pattern for 30 marks External Semester End Practical Examination:

Duration - This examination shall be of **Two hours** duration in laboratory

Sr. No.		Total 30 Marks
1	Experimental performance	20 Marks
2	Journal	05 Marks
3	VIVA	05 Marks

Paper Pattern for 20 marks Internal Semester End Practical Examination:

Duration - This examination shall be of **One hour** duration in laboratory

Sr. No.		Total 20 Marks
1	Practical/Theoretical/ Combined Performance	15 Marks
2	Overall Performance	05 Marks

Sem. – II

Syllabus

B.Sc. (Physics)

(Sem.- II)

Name of the Course: F.Y.B.Sc. SEM-II- Paper – I: Optics

Sr.No.	Heading	Particulars
1	Description the course : Including but Not limited to:	Introduction, relevance, Usefulness, Application, interest, connection with other courses, demand in the industry, job prospects etc
2	Vertical :	Major/Minor/Open Elective /Skill Enhancement / Ability Enhancement/Indian Knowledge System
3	Type :	Theory / Practical
4	Credits :	2 credits (1 credit = 15 Hours for Theory or 30 Hours of Practical work in a semester)
5	Hours Allotted :	30 Hours
6	Marks Allotted:	50 Marks
7	Course Objectives (CO): After successful completion of this course students will be able to: CO 1. Explain the nomenclature used in lenses, lens equations for single convex lenses, and sign convention. lens maker's equation, Newton's lens equation and principal foci positions. CO 2. Describe Lateral, Longitudinal and Angular magnification, Equivalent focal length and power of two thin lenses, Concept of cardinal points and their significance CO 3. Explain qualitatively Spherical aberration & reduction, chromatic aberration & reduction. CO 4. Study of Fresnel and Fraunhofer type of diffraction and Fraunhofer diffraction pattern due to a single slit and double slit, CO 5. Learn Michelson's Interferometer and its Applications CO 6. Describe Polarization and types of Polarization CO 7. The students learn to apply their knowledge to solve problems that are covered in the all syllabus.	
8	Course Outcomes (OC): After successful completion of this course students will be able to: OC 1. Understand the nomenclature used in lenses, lens equations for single convex lenses, and sign convention. lens maker's equation, Newton's lens equation and principal foci positions. OC 2. To Understand Lateral, Longitudinal and Angular magnification, Equivalent focal length and power of two thin lenses, Concept of cardinal points and their significance OC 3. To comprehend qualitatively Spherical aberration & reduction, chromatic aberration & reduction. OC 4. To understand Fresnel and Fraunhofer type of diffraction and Fraunhofer diffraction pattern due to a single slit and double slit, OC 5. To understand Michelson's Interferometer and its Applications	

	OC 6.To understand Polarization and types of Polarization OC 7.Students should be able to solve problems related to the topics that are covered in the syllabus.
9	Paper – I: Optics (30 Hrs) UNIT-I (15 Hrs) 1. Lenses and Lens Maker’s Equation: Introduction to lenses, Terminology and sign conventions, Introduction to Thin lenses and Lens equation for single convex lens, Lens maker’s equation: Positions of the Principal Foci and Newton’s Lens equation. (SBA: 4.1, 4.2, 4.3, 4.7, 4.8, 4.9, 4.10, 4.10.1, 4.11) 2. Magnification by a lens and power of lens: Lateral, Longitudinal and Angular magnification, Deviation by a thin lens and its power, Equivalent focal length of two thin lenses, Focal length of the equivalent lens & power of two thin lenses, Concept of cardinal points and their significance (SBA: 4.12, 4.12.1, 4.12.2, 4.12.3, 4.15, 4.16, 4.17, 4.17.1, 4.17.2, 4.17.3, 4.17.4, 5.2) 3. Introduction to Aberration in lenses: Spherical aberration & reduction, chromatic aberration & reduction (Qualitative). SBA: 9.2, 9.5, 9.5.1, 9.10 Suitable numerical with appropriate difficulty level. UNIT-II (15 Hrs) 1. Fresnel diffraction: Introduction, Huygens-Fresnel’s theory, Fresnel’s assumptions, Distinction between interference and diffraction, Fresnel and Fraunhofer types of diffraction, (SBA: 17.1, 17.2, 17.3, 17.6, 17.7) 2. Fraunhofer diffraction: Introduction, Fraunhofer diffraction at a single slit, intensity distribution in diffraction pattern due to a single slit, Fraunhofer diffraction at double slit (Qualitative), Distinction between single slit and double slit diffraction patterns. (SBA: 18.1, 18.2, 18.2.1, 18.4, 18.4.2) 3. Michelson’s Interferometer: Principle, construction, working, Applications of Michelson Interferometer: a) Measurement of wavelength b) Determination of the difference in the wavelength of two waves c) Determination of the refractive index of gases. (SBA: 15.7, 15.7.1 to 15.7.3, 15.8, 15.8.1, 15.8.2, 15.8.4) 4. Polarization: Introduction, Polarization, Types of Polarization (SBA: 20.1, 20.2, 20.5,20.5.1, 20.5.2, 20.5.3)
10	Text Books
11	Reference Books Dr. N. Subrahmanyam, Brijlal, and Dr. M. N. Avadhanulu, A Textbook of Optics, 25th Revised Edition (2012) S. Chand.

Internal Continuous Assessment: 40% (20 Marks)	Semester End Examination: 60% (30 Marks)
Continuous Evaluation through: Quizzes, Class Tests, presentation, project, role play, creative writing, assignment etc.(at least 3)	
Format of Question Paper: 30 Marks Duration: ONE Hour	
Unit -I (15Marks)	Q:1 A) Attempt any Two 10Marks v) Theory vi) Theory vii) Theory viii) Theory B) Attempt any One 05 Marks iii) Problem iv) Problrm
Unit -II (15Marks)	Q:2 A) Attempt any Two 10Marks v) Theory vi) Theory vii) Theory viii) Theory B) Attempt any One 05 Marks ii) Problem ii) Problrm

Name of the Course: F.Y.B.Sc. SEM-II, Paper-II: Fundamentals of Electronics

Sr.No.	Heading	Particulars
1	Description the course : Including but Not limited to:	Introduction, relevance, Usefulness, Application, interest, connection with other courses, demand in the industry, job prospects etc
2	Vertical :	Major/Minor/Open Elective /Skill Enhancement / Ability Enhancement/Indian Knowledge System
3	Type :	Theory / Practical
4	Credits :	2 credits (1 credit = 15 Hours for Theory or 30 Hours of Practical work in a semester)
5	Hours Allotted :	30 Hours
6	Marks Allotted:	50 Marks
7	Course Objectives(CO): After successful completion of this course students will be able to: CO 1. Explain the concepts of DC power supply and familiarize with diode and Zener diode circuits and its applications. CO 2. Describe DC transistors Biasing and α , β (dc and ac) gain, Inherent Variations of transistor Parameters and Stabilization, CO 3. Explain qualitatively methods of Transistor Biasing, Base Resistor or fixed bias, Emitter Bias and Voltage Divider Bias Methods. CO 4. Describe concepts of Number Systems and convert the numbers from one system to another. CO 5. Design and explain NAND, NOR and Ex-OR gate using basic gates, including their symbols and truth table. CO 6. Design logical circuit using basic gates and its applications CO 7. Work through problems pertaining to the topics covered in the syllabus.	
8	Course Outcomes (OC): After successful completion of this course students will be able to: OC 1. Comprehend the concepts of DC power supply and familiarize with diode and Zener diode circuits and its applications. OC 2. Understand DC transistors Biasing and α , β (dc and ac) gain, Inherent Variations of transistor Parameters and Stabilization, OC 3. Comprehend qualitatively methods of Transistor Biasing, Base Resistor or fixed bias, Emitter Bias and Voltage Divider Bias Methods. OC 4. Demonstrate the ability to convert from one number system to another OC 5. Understand Derived Gates NAND, NOR and Ex-OR gate, including their symbols and truth table. OC 6. Apply the knowledge to design logical circuit using basic gates and its applications	

	OC 7. Students must be able to work through problems pertaining to the topics covered in the syllabus.
9	Paper-II: Fundamentals of Electronics (30 Hrs) UNIT-I (15 Hrs) 1. DC Power Supply: Block diagram of a dc power supply – concept of a transformer, (Review: Half wave rectifier, Full wave rectifier) Bridge rectifier, PIV, Efficiency and Ripple factor of full wave rectifier, Capacitor Filter, Need for voltage regulation - Zener diode as voltage stabilizer, Clipper and Clampers (Basic diode based circuits only). (BN: 1.15, 2.6, 2.7, 2.8, 2.9, 2.10, 15.2, 15.3 AD: 4.2, 22.1 2.) 2. Transistor dc Biasing: (Review: transistor structure and characteristics), Definition of gains α, β (dc and ac) and relation between them, load line analysis, operating point, cut-off and saturation points, Inherent Variations of transistor Parameters, Stabilization, Necessity of a Transistor Biasing Circuit, Stability Factor, Methods of Transistor Biasing, Base Resistor or fixed bias, Emitter Bias and Voltage Divider Bias Methods (Qualitative Analysis only, No mathematical derivation), Stability factor for Potential Divider Bias. (BN: 3.1, 3.2, 3.3, 3.4, 3.5, 3.6, 3.7, 4.1, 4.2) UNIT-II (15 Hrs) 1. Number Systems: Binary number system: Binary to decimal and Decimal to binary conversion, Hexadecimal number system: Hexadecimal to decimal Conversion, Decimal to hexadecimal conversion, Hexadecimal to binary conversion, Binary to hexadecimal conversion. (LMS: 5.1 to 5.5 2.) 2. Derived Gates: (Review: Basic Logic gates), NAND and NOR as Universal Building blocks, Ex-OR gate: logic expression, logic symbol, truth table, Implementation using basic gates and its applications – Parity generator and checker, Half adder and Full adder. (LMS: 2.1, 2.2 Tokheim: 3.6, 3.8, 10.2, 10.3 3.) 3. Boolean Algebra: Boolean theorems, De-Morgan theorems, Sum of Product (SOP) and Product of sum (POS) methods, Simplification of logical expressions. (LMS: 3.1, 3.2, 3.7, 3.8)
10	Text Books
11	Reference Books 1. BN: R. L. Boylestad and L. Nashelsky, Electronic devices and Circuit Theory - 10th Edition, Pearson 2. LMS: Leach, Malvino, Saha, Digital Principles and Applications – 6 th Edition. Tata McGraw Hill

Internal Continuous Assessment: 40% (20 Marks)	Semester End Examination: 60% (30 Marks)
Continuous Evaluation through: Quizzes, Class Tests, presentation, project, role play, creative writing, assignment etc.(at least 3)	
Format of Question Paper: 30 Marks Duration: One Hour	
Unit -I (15Marks)	Q:1 A) Attempt any Two 10Marks i) Theory ii) Theory iii) Theory iv) Theory B) Attempt any One 05 Marks i) Problem ii) Problrm
Unit -II (15Marks)	Q:2 A) Attempt any Two 10Marks i) Theory ii) Theory iii) Theory iv) Theory B) Attempt any One 05 Marks i) Problem ii) Problrm

SEMESTER - II
PHYSICS PRACTICAL COURSE –USPHP2

INSTRUCTIONS:

- 1) All the measurements and readings should be written with proper units in SI system only.
- 2) After completing all the required number of experiments in the semester and recording them in journal, student will have to get their journal certified and produce the certified journal at the time of practical examination.
- 3) While evaluating practical, weightage should be given to circuit/ray diagram, observations, tabular representation, experimental skills and procedure, graph, calculation and result.
- 4) Skill of doing the experiment and understanding physics concepts should be more important than the accuracy of final result.

Note: Exemption of two experiments from section A and / or B and / or C may be given if student carries out any one of the following activities.

- Collect the information of at least five Physicists with their work or any three events on physics, report that in journal.
- Execute a mini project to the satisfaction of teacher in-charge of practical.
- Participate in a study tour or visit & submit a study tour report.
- For practical examinations, the learner will be examined in **ONE** experiment (one from any group).
- Each experiment will be of **TWO** lecture hours' duration.
- A Minimum 4 from each group and in all minimum 8 experiments must be reported in journal.

All the skill experiments are required to be completed compulsorily. Students are required to report all these experiments in the journal. Evaluation in viva voce will be based on regular experiments and skill experiments.

A learner will be allowed to appear for the semester and practical examination only if he submits a certified journal of Physics or a certificate that the learner has completed the practical course of Physics Semester II as per the minimum requirements.

A. Regular Experiment:

Sr No	Name of the Experiments
GROUP A	
1	Study of LASER Beam Divergence
2	Spectrometer: To determine of angle of Prism
3	Spectrometer: To determine refractive index of prism material
4	Combination of Lenses: To determine equivalent focal length of a lens system by magnification method
5	Newton's Rings: To determine radius of curvature of a given convex lens using Newton's rings.
6	Determination of diameter of thin wire using Wedge Shaped Film
GROUP B	
7	Study of Logic gates & To verify De Morgan's Theorems
8	To study EX-OR Gate and verify its truth table
9	To study half adder and full adder and verify their truth table Ex-OR Gate
10	To study load regulation of a Bridge Rectifier
11	To study Zener Diode as Regulator
12	Transistor configurations : CB/CE/CC (study of input-output characteristics)
GROUP C: DEMONSTRATION EXPERIMENT	
1	Radius of ball bearings (single pan balance)
2	Use of Oscilloscope: Wave forms at output of half wave , bridge rectifiers with and without Capacitor filter, Ripple
3	Use of PC for graph plotting
4	I-V Characteristics of LED
5	Testing of components (Resistors , Diode , Transistor , capacitor)
6	Study of I-V characteristics of solar cell

Note: Minimum **8** experiments (Four From each group) and **4** Demo experiments should be completed and reported in the journal, in the first semester. **Certified Journal is a must**, to be eligible to appear for the semester end practical examination.

Semester End Practical Examination:

Semester End Practical Examination: 30 Marks External & 20 Marks Internal

Scheme of Examination: 30 Marks Duration: TWO Hours

There will be internal 20 Marks assessment for practical. A candidate will be allowed to appear for the semester end practical examination only if the candidate submits a certified journal at the time of practical examination of the semester or a certificate from the Head of the Department /Institute to the effect that the candidate has completed the practical course of that semester of F.Y.B.Sc. Physics as per the minimum requirement. The duration of the practical examination will be **TWO** hours. There will be **ONE** experiment (from any group) through which the candidate will be examined in practical. The questions on slips for the same should be framed in such a way that candidate will be able to complete the task and should be evaluated for its skill and understanding of physics.

F.Y.B.Sc. Physics Evaluation pattern and Question paper pattern for Semester End Practical Examination of Major Course

Internal Continuous Assessment: 40% (20 Marks)	Duration of semester End examination (Internal)
Viva/ assignment/ objective question test (15 Marks), Overall performance (5 Marks) = 20 Marks	1 Hr

External Semester End Examination: 60% (30 Marks)	Duration of semester End examination (External)
As per paper pattern	2 Hr

Continuous Evaluation through: Practical

Format of Question Paper:

Major Practical: 2 credits

Practical of 2 credits, Duration: 60 Hrs, Total marks: 50

(30 Marks for External Practical Exam + 20 Marks for Internal Practical Exam.)

Paper Pattern for 30 marks External Semester End Practical Examination:

Duration - This examination shall be of **Two hours** duration in laboratory

Sr. No.		Total 30 Marks
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1	Experimental performance	20 Marks
2	Journal	05 Marks
3	VIVA	05 Marks

Paper Pattern for 20 marks Internal Semester End Practical Examination:

Duration - This examination shall be of **One hour** duration in laboratory

Sr. No.		Total 20 Marks
1	Practical/Theoretical/ Combined Performance	15 Marks
2	Overall Performance	05 Marks

SEMESTER-II
Physics (Vocational Skill Course)
Course code: USPHPVSC2
Title: - PCB Making

Learning Objectives:

1. Learn techniques required for soldering of electronic components.
2. Learn to create effective PCB layouts.

Learning Outcomes:

On successful completion of this course students will be able to:

1. Solder basic electronic components on a PCB.
2. Develop schematic electronic circuit designing skills.
3. Develop PCBs.

Unit I: Circuit Prototyping Skills

(15 Hours)

1. Soldering electronic components
2. Circuit assembly on general purpose board
 - A. Bridge rectifier with capacitor filter and regulator
 - B. Second order active Filter
 - C. Wheatstone bridge for temperature measurement
 - D. 555 based LED flasher
3. Testing
4. Soldering of surface mount devices

Unit II: PCB Design Software

(15 Periods)

1. Schematic circuit entry software
2. PCB Layout
 - A. Footprint assignment
Creating board outlines for various layers: Bottom routing, Solder mask bottom, Legend or silk screen Top, Drill layer
 - B. Placement on the board
 - C. Routing
 - D. Post processing, Assembly and testing

List of Experiments:

1. Single side PCB
2. Double side PCB
3. Multi-layer PCB
4. Schematic / Components design review.
5. Determine what footprints are required to be built, and build them incorporating.
6. Component placement.
7. Power and Ground Plane assignment.
8. Critical net routing.

Reference:

1. Electronic Product Design Vol. I Basic for PCB Design – by Er. Mehta S.D.
2. Printed circuit boards: Design fabrication , Assmbly and testing – R S Khandpur.
3. PCB Design and Layout Fundamentals – by Roger Hu
4. PCB Design and Technology – by Walter C Bosehart

Internal Continuous Assessment: 40% (20 Marks)	Semester End Examination: 60% (30 Marks)
Continuous Evaluation through: Practical's	

Format of Question Paper:**VSC: 2 credits****VSC of 2 credits, Duration: 30 Hrs , Total marks: 50****(30 Marks for Theory paper + 20 Marks for Practical Exam.)**

Semester End Theory Examination: 60% (30 Marks)	Internal Semester End Practical Examination: 40 % (20 Marks)
As per paper pattern attached	As per practical exam. pattern attached

F.Y.B. Sc. Physics Evaluation pattern and Question paper pattern for Semester End VSC Practical Examination

Internal Continuous Assessment: 40% (20 Marks)	Duration of semester End examination (Internal)
Viva/ assignment/ objective question test (15 Marks), Overall performance (5 Marks) = 20 Marks	1 Hr
External Semester End Examination: 60% (30 Marks)	Duration of semester End examination (External)
As per paper pattern	2 Hr

Continuous Evaluation through: Practical

Format of Question Paper:

Major Practical: 2 credits

Practical of 2 credits, Duration: 30 Hrs , Total marks: 50

(30 Marks for External Practical Exam + 20 Marks for Internal Practical Exam.)

Paper Pattern for 30 marks External Semester End Practical Examination:

Duration - This examination shall be of **Two hours** duration in laboratory

Sr. No.		Total 30 Marks
1	Experimental performance	20 Marks
2	Journal	05 Marks
3	VIVA	05 Marks

Paper Pattern for 20 marks Internal Semester End Practical Examination:

Duration - This examination shall be of **One hour** duration in laboratory

Sr. No.		Total 20 Marks
1	Practical/Theoretical/ Combined Performance	15 Marks
2	Overall Performance	05 Marks

Physics (Skill Enhancement Course) - 2 credits

Course code: USPHSEC2

Title: - ICT tools for Physics

Learning Objectives:

1. To know how to use the most common Microsoft Office programs.
2. To be able to create documents for printing and sharing.
3. To be able to create and share presentations.
4. To be able to manage and store data in a spreadsheet.

Learning Outcomes:

On successful completion of this course students will be able to

1. Create a word document, save the word document and print the word document.
2. Demonstrate various insert features and mail merge feature of a word document.
3. Create, edit, save, format and print presentations.
4. Create and manipulate simple slide shows.
5. Create, open, view, edit, save and print a workbook.
6. Learn to use functions, formulas, charts and graphs.

Unit 1: MS-Word and MS

(30 Hours)

1. Introduction to MS Word, starting word, creating a Document, Saving and Printing a document, Move and Copy Text, Cut and Paste, Finding Text, Replace Command, Checking Spelling and Grammar
2. Inserting Picture, Formatting Text, fonts, Using Bullets and Numbering in Paragraphs, Inserting equations and symbols
3. Page Setup, Inserting Page Breaks, Using Headers and Footers in the Document, inserting page numbering, Print Preview, Print Options
4. Creating Tables, Formatting a Table
5. Using Mail Merge.
6. Introduction to power point presentation, creating a blank presentation, adding new slides, saving a presentation, printing options
7. Designing a presentation
8. Animation and transition, Slide show

Unit 2: MS-EXCEL

(30 Hours)

1. Creating spreadsheet and adding information to it, different data types
2. Moving data values, editing data values, inserting/ deleting rows and columns
3. Data editing: find & replace, spell check,
4. Data formatting techniques
5. Working with different mathematical, text, date and time formulae
6. Page layout options, adding header, footer and page numbering
7. Working with sort and filter functions
8. Working with multiple spreadsheets.

All the above topics to be covered through Hands on sessions.

Reference:

1. *Mastering MS Office (ebook) by Bittu Kumar*
2. *Excel 2010 Bible by John Walkenbach, John Wiley & Sons*

F.Y.B. Sc. Physics Evaluation pattern and Question paper pattern for Semester End SEC Practical Examination

Internal Continuous Assessment: 40% (20 Marks)	Duration of semester End examination (Internal)
Viva/ assignment/ objective question test (15 Marks), Overall performance (5 Marks) = 20 Marks	1 Hr

External Semester End Examination: 60% (30 Marks)	Duration of semester End examination (External)
As per paper pattern	2 Hr

Continuous Evaluation through: Practical

Format of Question Paper:

Major Practical: 2 credits

Practical of 2 credits, Duration: 60 Hrs , Total marks: 50

(30 Marks for External Practical Exam + 20 Marks for Internal Practical Exam.)

Paper Pattern for 30 marks External Semester End Practical Examination:

Duration - This examination shall be of **Two hours** duration in laboratory

Sr. No.		Total 30 Marks
1	Experimental performance	20 Marks
2	Journal	05 Marks
3	VIVA	05 Marks

Paper Pattern for 20 marks Internal Semester End Practical Examination:

Duration - This examination shall be of **One hour** duration in laboratory

Sr. No.		Total 20 Marks
1	Practical/Theoretical/ Combined Performance	15 Marks
2	Overall Performance	05 Marks

Letter Grades and Grade Points:

Semester GPA/ Programme CGPA Semester/ Programme	% of Marks	Alpha-Sign/ Letter Grade Result	Grading Point
9.00 - 10.00	90.0 - 100	O (Outstanding)	10
8.00 - < 9.00	80.0 - < 90.0	A+ (Excellent)	9
7.00 - < 8.00	70.0 - < 80.0	A (Very Good)	8
6.00 - < 7.00	60.0 - < 70.0	B+ (Good)	7
5.50 - < 6.00	55.0 - < 60.0	B (Above Average)	6
5.00 - < 5.50	50.0 - < 55.0	C (Average)	5
4.00 - < 5.00	40.0 - < 50.0	P (Pass)	4
Below 4.00	Below 40.0	F (Fail)	0
Ab (Absent)	-	Ab (Absent)	0

Sd/-

**Sign of the BOS Chairman
Dr. T.N. Ghorude
Board of Studies in
Physics**

Sd/-

**Sign of the
Offg. Associate Dean
Dr. Madhav R. Rajwade
Faculty of Science &
Technology**

Sd/-

**Sign of the
Offg. Dean
Prof. Shivram S. Garje
Faculty of Science &
Technology**